

TagAlong: Scaling Scene Text Recognition for Short-Form Video

Team #TagAlong

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May 4, 2026

Abstract

This report presents the final milestone of TagAlong, a scene text recognition project motivated by the goal of extracting structured information from short-form travel videos for downstream itinerary generation. Building on the Milestone 1 CRNN baseline (11.7% test word accuracy) and the Milestone 2 architecture comparison (best: 46.7% test on IC13+IC15), Milestone 3 isolates the effect of training-data scale by retraining the M2 best model on a combined IC13+IC15+IC19 dataset of 41,650 samples — approximately 9× the M2 training set. The scaled model achieves **84.5% validation and 73.1% test word accuracy**, a 6× improvement over the M1 baseline and a 1.5× improvement over the M2 best, confirming the M2 hypothesis that the model was data-limited rather than capacity-limited. The detection and audio branches outlined in the M2 plan were deferred to future work due to compute and time constraints; this report focuses on the recognition component and the data-scaling result.

1 Introduction

Problem Definition

Short-form social media content (Instagram Reels, TikTok, YouTube Shorts) embeds dense structured information in visual text overlays, on-screen captions, location tags, and storefront signage. Extracting this text reliably from individual video frames is a prerequisite for any downstream system that converts unstructured travel content into structured itineraries, recommendations, or maps. TagAlong targets this recognition problem at the still-image level: given a word-region image, output its text transcription.

Motivation

Existing scene text recognition systems are trained primarily on document-style benchmarks or curated competition datasets (ICDAR 2013/2015) with limited distributional coverage. Travel-reel content typically combines stylized fonts, low-contrast overlays, and arbitrary text shapes — distributional regions where smaller benchmarks are sparse. The TagAlong project investigates whether expanding training data into more diverse public scene-text datasets (notably ICDAR 2019-ArT, which contains arbitrary-shape Latin text) yields meaningful gains over a strong CRNN baseline trained on smaller competition data alone.

Objectives (Milestone 3)

- Scale the M2 best recognition model to a substantially larger training distribution (IC13 + IC15 + IC19) and quantify the gain in word accuracy and character-level F1.
- Test the M2 hypothesis that the Improved CRNN was data-limited rather than capacity-limited by holding architecture fixed and varying only the training set.

- Defer the detection and audio components originally outlined in the M2 plan to future work, given the compute and time cost of training recognition models on the expanded dataset (see §7).

2 Related Work

The CRNN architecture remains the standard baseline for scene text recognition, combining a convolutional feature extractor, a bidirectional recurrent sequence model, and a CTC decoder. M2 compared CRNN against two attention-based alternatives: SAR, which uses 2D attention over the feature map, and DTrOCR, a decoder-only transformer. M2 found that on IC13-scale data both attention-based models were severely under-trained, while the Improved CRNN converged reliably; M3 holds that finding and asks whether further data closes the gap to published benchmark numbers without changing architecture.

The ICDAR Robust Reading Competition series (2013, 2015, 2017-MLT, 2019-ArT) provides the public benchmarks used in this work. IC13 contains horizontal scene text with deliberate framing; IC15 contains incidental scene text with perspective distortion; IC19-ArT introduces arbitrarily-shaped text including curved and rotated instances. Each dataset uses a different annotation convention (per-line ground-truth files for IC13, per-image polygons for IC15, JSON for IC19), which required separate loader paths in the M3 training driver.

3 Methodology

Overall Approach

Milestone 3 retrains the M2 best model — the Improved CRNN with beam-search CTC decoding — on a larger combined dataset. No architectural changes are introduced. This isolates the effect of training-data scale and tests the M2 conclusion that the model was data-limited.

The training pipeline follows M2 exactly: 32×256 RGB input, ImageNet-statistic normalization, ColorJitter + RandomAffine + GaussianBlur augmentation at training time only, AdamW optimization with cosine LR schedule, label smoothing not applied (CRNN uses CTC loss), and beam-search decoding with width 5 at inference.

3.1 Design Decisions

- No architectural changes from M2. The Improved CRNN (5-block CNN with BatchNorm and LeakyReLU, height-1 feature collapse, 2-layer Bi-LSTM with LayerNorm, dropout 0.3) was the M2 best performer. M3 holds it fixed and varies only training data so that any observed gain is attributable to data rather than capacity.
- IC17-MLT was excluded due to download size and training-time cost (commit [1c605d05](#)). IC17's multilingual scope and dataset volume made the marginal gain unjustifiable within the milestone timeline; IC19-ArT alone provided sufficient distributional diversity (arbitrary-shape Latin text) to test the data-scaling hypothesis.
- ≤25-character annotation filter for IC19 (commit [f2b377da](#)). IC19 contains phrase-level annotations whose length, after the CRNN's CNN downsampling, can exceed the available CTC time-step budget T . Annotations longer than 25 characters were filtered at dataset-load time to prevent training-time CTC errors.

- IC15 sample count differs from M2 (4,468 → 4,060). The ≤ 25 -char filter introduced for IC19 was applied uniformly across all M3 training data for consistency, removing some longer IC15 annotations that M2 had retained.
- Latin-only filter on IC19. IC19-ArT contains both Latin and non-Latin (Chinese, Arabic) text instances. Non-Latin samples were excluded at load time because the M3 vocabulary covers only the 37 alphanumeric classes inherited from M1/M2. Multilingual support is identified as future work (see §11).

4 Implementation

Technologies

PyTorch 2.10.0, torchvision, editdistance, matplotlib. The M3 stack is intentionally a strict subset of the M2 stack — torchaudio, OpenCV, and ffmpeg, which were planned for the audio and video-frame branches, were not used in this milestone.

System Components

- `src/scale_experiment.py` — multi-dataset training driver. Constructs combined train and validation loaders from IC13, IC15, and IC19, applies the ≤ 25 -char filter and Latin-only filter, and runs the standard M2 training loop.
- `src/datasets/` — three loader classes (IC13Dataset, IC15Dataset, IC19ArTDataset) handling per-dataset annotation conventions: per-line gt.txt for IC13, per-image polygons for IC15, and JSON for IC19.
- `models/improved_crnn.py` — unchanged from M2. 5-block CNN (3→64→128→256→256→512) with BatchNorm and LeakyReLU(0.1) in every block, final 2×1 convolution collapsing feature height to 1, two Bi-LSTM layers (hidden 256) with LayerNorm and dropout 0.3, linear projection to 37 CTC classes.
- `prepare_ic15.py` — IC15 preprocessing carried from M2. Applies a PIL QUAD perspective transform to warp tilted text crops into axis-aligned 32×256 rectangles.

Final Training Set Composition

Dataset	Samples	Notes
ICDAR 2013	675	Carried from M2
ICDAR 2015	4,060	Perspective-warped via <code>prepare_ic15.py</code>
ICDAR 2019-ArT	36,915	Latin-only, ≤ 25 char filter applied
Total	41,650	~9× M2's IC13+IC15 size

Table 1: M3 combined training set. IC19-ArT dominates the distribution (~89% of samples).

5 Experiments and Evaluation

Experimental Setup

Hyperparameter	Value
Epochs	20 (scale experiment)
Batch size	32
Optimizer	AdamW
Learning rate	3×10^{-4}
LR schedule	Cosine annealing
Weight decay	1×10^{-4}
Gradient clip	5.0 (max norm)
Beam width (CTC decode)	5
Image size	32×256 px
Validation split	20% of IC13 train (seed 42)
Vocabulary	37 classes (a-z, 0-9, CTC blank)

Table 2: M3 training hyperparameters. Architecture and training loop are unchanged from the M2 Improved CRNN configuration; only the training data composition differs.

Metrics

- Word accuracy: fraction of predictions exactly matching ground truth (case-insensitive, alphanumeric only).
- Character-level F1: average character-level precision/recall F1 across the test set.
- CTC validation loss: tracked during training to verify convergence.

Baselines

Two baselines are carried forward from previous milestones for comparison: the M1 CRNN trained on IC13 only (611 samples, 11.7% test) and the M2 best model — the same Improved CRNN architecture used in M3 — trained on IC13+IC15 (~4,700 samples, 46.7% test). The M3 result isolates the contribution of adding IC19 to the training set.

5.1 Results

Training Data	Samples	Val Acc	Test Acc
IC13 only (M1 baseline)	611	21.1%	11.7%
IC13 + IC15 (M2 Exp 5)	~4,700	54.2%	46.7%
IC13 + IC15 + IC19 (M3)	41,650	84.5%	73.1%

Table 3: Word accuracy across data-scaling configurations. M3 best in bold. Architecture is held fixed across the M2 Exp 5 and M3 rows; only the training set differs.

5.2 Discussion

The IC13+IC15 → IC13+IC15+IC19 step added approximately 36,900 training samples and yielded a 30.3 percentage-point gain in validation accuracy (54.2% → 84.5%) and 26.4 percentage-point gain in test accuracy (46.7% → 73.1%). The earlier IC13 → IC13+IC15 step in M2 added approximately 3,400 samples for a 7.5-point test gain. The returns scale roughly proportionally to the logarithm of samples added, which is consistent with standard data-scaling behavior reported in the recognition literature and supports the M2 hypothesis that the model was data-limited rather than capacity-limited.

The validation–test gap (84.5% vs 73.1%, an 11.4-point spread) is larger in absolute terms than M2's gap (54.2% vs 46.7%, 7.5 points) but proportionally similar relative to the higher accuracy band. The IC13 test set is known to differ distributionally from the IC13 training data (deliberate vs. partially-incidental framing), and this gap reflects that distribution shift rather than overfitting in the training-curve sense — training and validation curves on the M3 combined set tracked closely throughout training.

IC19-ArT drove the bulk of the gain. With 36,915 samples it constitutes approximately 89% of the M3 training set, and the arbitrary-shape Latin text it contains is distributionally different from the deliberately-framed IC13 and incidentally-captured IC15 imagery. Adding this third regime of scene text — curved and rotated instances — broadened the visual feature distribution the CRNN learned without changing the recognition task itself.

6 Improvements over M2

- Larger training distribution. M2's best configuration used IC13+IC15 (~4,700 samples). M3 expands this to IC13+IC15+IC19 (41,650 samples, ~9× larger), introducing IC19-ArT's arbitrary-shape Latin text as a third distributional regime.
- Confirmed M2's data-limited hypothesis. M2 concluded that the Improved CRNN's residual error on IC13+IC15 was driven by data scarcity rather than model capacity, and M3 was designed as a direct test of that claim. The 26.4-point test-accuracy gain at fixed architecture supports the hypothesis and isolates data scale as the binding constraint at the M2 operating point.
- Uniform annotation filtering. The ≤25-character filter introduced for IC19 was applied uniformly across all training data, including IC15. This produced a consistent CTC label-length distribution across datasets and a small reduction in IC15 sample count (4,468 → 4,060) relative to M2.

7 Changes from the M2 Plan

The M2 "Next Steps" section outlined three M3 directions: (a) further dataset scaling and tuning of the recognition model, (b) operating on full scene images via a text detection component, and (c) integrating recognition with downstream video-frame and audio-narration processing toward an end-to-end reel-to-itinerary pipeline. Given the compute cost of training recognition models on the expanded dataset and the milestone time budget, M3 prioritized (a) and deferred (b) and (c) to future work.

This was a deliberate scope decision rather than an unplanned cut. The M2 conclusion that the model was data-limited identified data scaling as the highest-leverage single change available, and the 1.5× test-accuracy improvement over M2 best (46.7% → 73.1%) supports that

prioritization in retrospect. The detection and audio branches remain on the future-work list (see §11) and do not contradict any M2 finding; they are simply unbuilt.

The other deviation from the M2 plan was the exclusion of IC17-MLT, which had been mentioned as a candidate dataset. IC17's download size and training-time cost made it impractical to include alongside IC19 within the milestone window; IC19 alone was sufficient to test the data-scaling hypothesis, and the alphanumeric-only vocabulary used in M1/M2/M3 made IC17's multilingual content largely unusable without vocabulary expansion.

8 Challenges

- Training time on combined datasets. The IC13+IC15+IC19 set (~41,650 samples) at batch 32 over 20 epochs took substantially longer per run than the M2 IC13+IC15 configuration, which constrained how many configurations could be explored within the milestone window. This is the single largest reason the detection and audio branches outlined in M2 were deferred.
- CTC time-step budget on long annotations. IC19-ArT contains phrase-level annotations whose character count, after the CRNN's CNN downsampling, can exceed the available CTC output sequence length. Without filtering, this produces silent training-time errors where some labels cannot be aligned. Resolved by filtering annotations longer than 25 characters at dataset load time (commit f2b377da) and applying the same filter uniformly to IC15 for consistency.
- Annotation-format heterogeneity across IC datasets. IC13 uses per-line gt.txt, IC15 uses per-image polygon files, and IC19 uses a top-level JSON manifest. Three separate loader classes were required, each producing a uniform (image_tensor, label_indices) sample interface for the shared training loop.
- Latin / non-Latin filtering on IC19. IC19-ArT contains substantial non-Latin content (Chinese, Arabic) which is incompatible with the 37-class alphanumeric vocabulary inherited from M1/M2. Non-Latin samples are excluded at load time. This is documented as a vocabulary limitation in §9 rather than a defect; expanding the vocabulary would have required retraining all earlier baselines for fair comparison.

9 Limitations

- Recognition only — no detection. The system requires pre-cropped word images at inference and cannot process full scene images end-to-end. This is the largest gap relative to a deployable pipeline and is the first item on the future-work list.
- No multimodal evaluation. Travel reels carry information in narration and on-screen captions in addition to scene text. This work covers only the visual word-recognition component; the audio branch outlined in the M2 plan was not built.
- DTrOCR not retested at full scale. M2 evaluated DTrOCR on IC13 alone, where it underperformed the Improved CRNN due to data scarcity. Retraining DTrOCR on the full IC13+IC15+IC19 set was deferred for the same time-and-compute reasons that deferred the detection and audio branches; whether transformer-based recognition closes the gap to CRNN at this data scale therefore remains untested in this work.
- Vocabulary is alphanumeric-only (37 classes). No punctuation, no capitalization, no non-Latin scripts. All non-Latin annotations in IC19 were dropped. This restricts the system's applicability to the subset of reel content whose informative text falls within these classes.

- No evaluation on user-generated reels. All evaluation uses the IC13 test set. Performance on actual short-form video frames — with motion blur, compression artifacts, and arbitrary text styling beyond the ICDAR distribution — is unmeasured.
- Validation–test distribution gap. The 11.4-point spread between validation (84.5%) and test (73.1%) accuracy reflects a known distributional difference between the IC13 training and test splits and bounds the confidence with which test accuracy can be projected to other scene-text distributions.
- No real-time or mobile deployment. The system is research-grade Python; inference latency, memory footprint, and quantization are unaddressed.

10 Conclusion

TagAlong's three-milestone trajectory traced a clean architecture-then-data progression. M1 established a CRNN baseline (11.7% test accuracy) and identified severe overfitting driven by a 611-sample training set. M2 fixed the architecture — deeper CNN with BatchNorm, LayerNorm on the recurrent layers, beam-search decoding — and partially addressed the data problem by adding IC15 (46.7% test). M3 finishes the data fix by adding IC19-ArT and confirms the M2 hypothesis that the binding constraint at the M2 operating point was training-data scale, not model capacity.

The headline result is a 6× improvement in test word accuracy over the M1 baseline (11.7% → 73.1%) and a 1.5× improvement over the M2 best (46.7% → 73.1%), achieved with no architectural changes and no new modeling components — only a larger and more distributionally diverse training set. The detection, audio, and end-to-end reel-pipeline components originally outlined in the M2 plan are not built in this work and are documented as future-work items rather than completed contributions, which keeps the report's claims aligned with what was actually implemented and evaluated.

11 Future Work

1. Text detection. Replace the current pre-cropped-input assumption with a real detection front-end (DBNet, CRAFT, or EAST) so the system can operate on full scene images and full video frames.
2. Audio branch. Add a parallel speech-recognition path to capture place names and other information that travel-reel narration carries but on-screen overlays often omit. A wav2vec2 or Whisper-tiny backbone would be more appropriate than a from-scratch CRNN-style audio model.
3. Synthetic pretraining. Pretrain the CRNN on SynthText or MJSynth before fine-tuning on the IC datasets to test whether the M3 data-scaling gain can be matched or exceeded with synthetic data.
4. Vocabulary expansion. Add punctuation, capitalization, and at least one additional script (likely Chinese, given IC17/IC19 coverage) to broaden applicability to non-English reels.
5. End-to-end itinerary generation. Run named-entity recognition over the fused recognition output, resolve place mentions against a maps API, and produce ordered itineraries — the original project motivation.
6. DTrOCR at full scale. Retrain the M2 DTrOCR variant on IC13+IC15+IC19 to test whether transformer recognition closes the gap to CRNN at this data scale, controlling the comparison that M2 had to leave open due to data scarcity.

7. Mobile deployment. Wire the trained recognizer into the scaffolded Android/mobile project directories, with quantization and inference-latency benchmarking.

12 Reproducibility

Repository

<https://github.com/hsu-emily/TagAlong/tree/DeepLearning>

Dependencies

- Python 3.10+
- PyTorch 2.10.0
- torchvision
- editdistance
- matplotlib
- Full pinned list in `requirements.txt`.

Running the M3 Scale Experiment

```
python src/scale_experiment.py
```

This command builds the IC13+IC15+IC19 combined loader (with the ≤ 25 -char and Latin-only filters applied), trains the Improved CRNN with the hyperparameters listed in Table 2, and writes per-epoch metrics and a final results JSON to `results/m3/`.

Notebook

`TagAlong_Colab_M3.ipynb`

References

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Contribution Statement

Emily: scale experiment driver and training runs; Aileen: training runs, report writing; Lilymoon: training runs, and report writing.